

Thread - 2026-02-07

<https://x.com/RiikonenMarko/status/2020261208054456357>

1/9 - 2026-02-07 22:19 UTC

7/8 Feb 2026, Rovaniemi. The theme of the night was Moilanen arc. Took also crystal samples. Here is a single exposure from the beginning when Moila was at its best. Actually, I think it was the best I have seen. Typically, in the beam Moila is not conspicuous visually – you can't see any shape in it even if the shape comes out fine in a single image. But this one was a clear, solid V-shape also visually. Dynamic range of the display was challenging, though. Had leave anything outside 22° halo pretty dark in order not to overexpose Moila.

The crystals were small, so I don't expect any breakthroughs. But it seemed like I have made some progress in fixing the milky central area that has plagued my photos. I took off the condenser and judging by what I saw from the camera screen, the photos came out better.

This was from the backyard gunning in Rautiosaari area. I had to photograph in a small field surrounded by trees. The car showed -19 C on arrival, upon leaving -25 C. The display was still going when I left, but not nearly as good as in the beginning. HID first, then Olight Javelot Turbo.



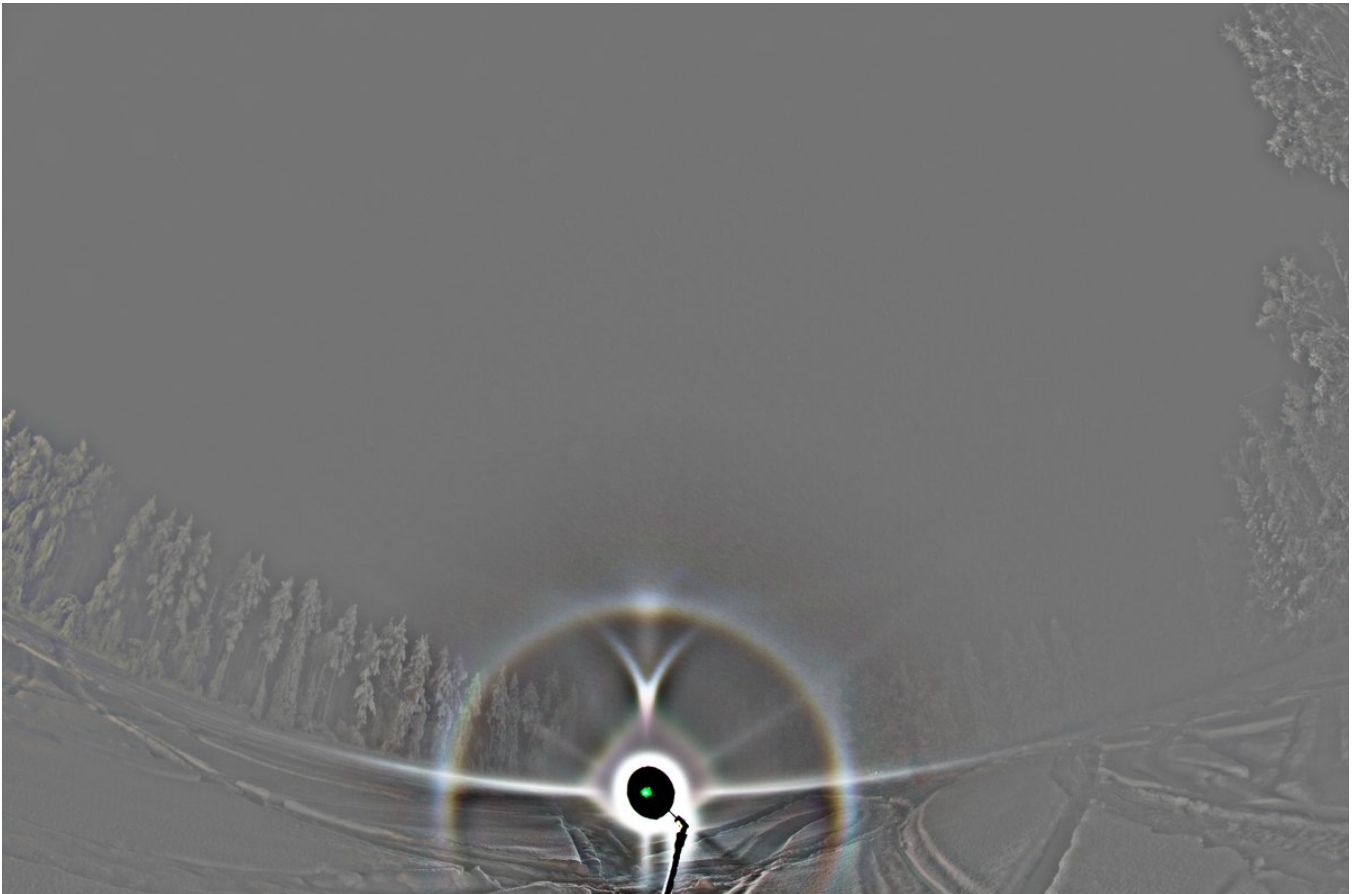
2/9 - 2026-02-08 09:25 UTC

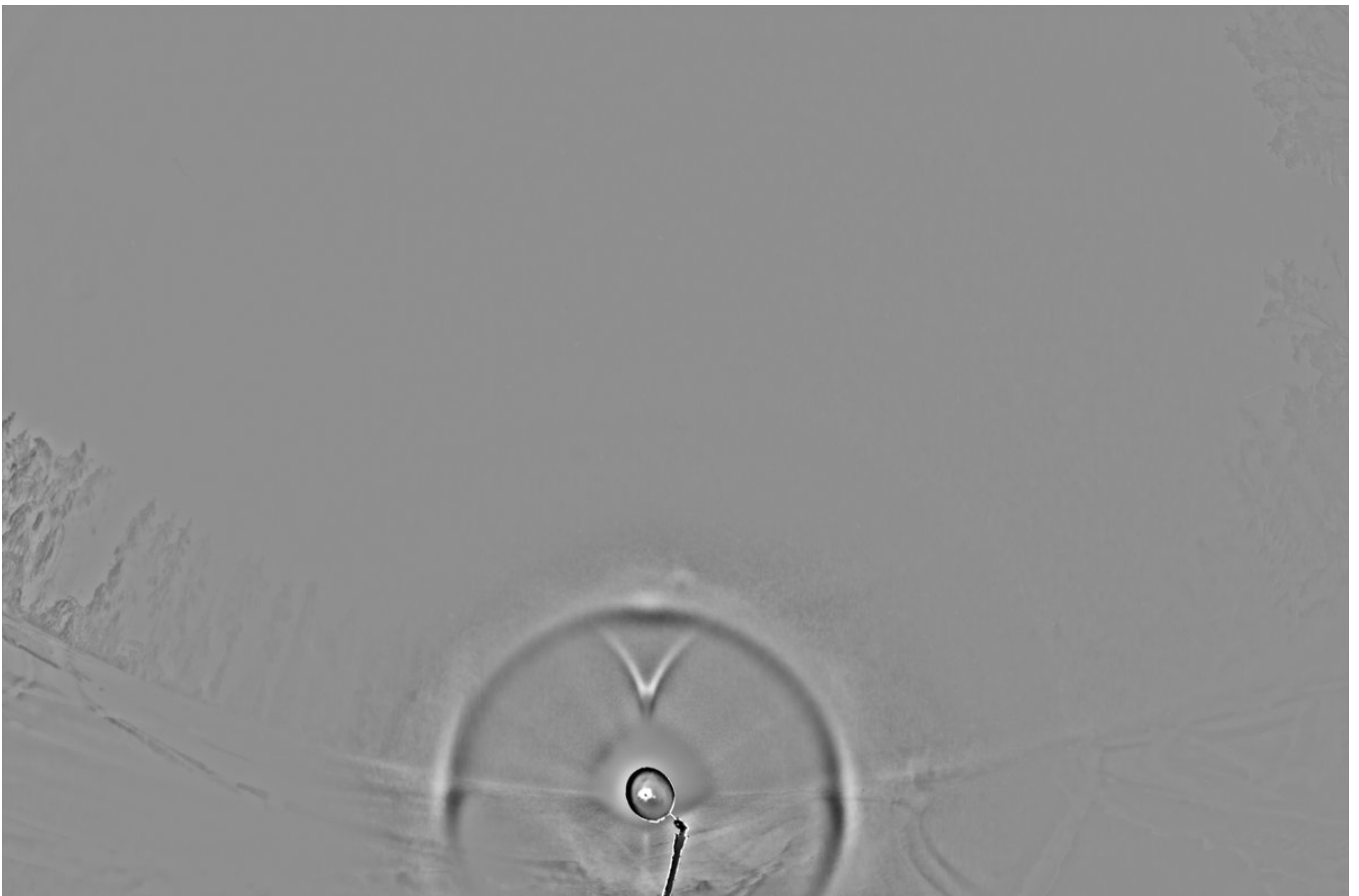
7/8 Feb 2026, Rovaniemi. 78 x 10s. The first half of the first set. HID about level with the camera. Stacking didn't really bring that much improvement. The swarm was so thick that already the single 10s exposure of the above post is quite good.

The beam was not pointing directly at the camera, it was shooting overhead, so the below-lamp part of

the display is underrepresented. The HID beam is not clean, it has spiky beam artefacts. They remain in b-r (second image) but are cleaned out by b-g (third image).

Still no Moilanen helic arc, but then the normal helic arc is neither seen below the lamp, possibly due to the underrepresentation of the below-lamp part. In the next post for reminder the Moilanen helic arc simulated.

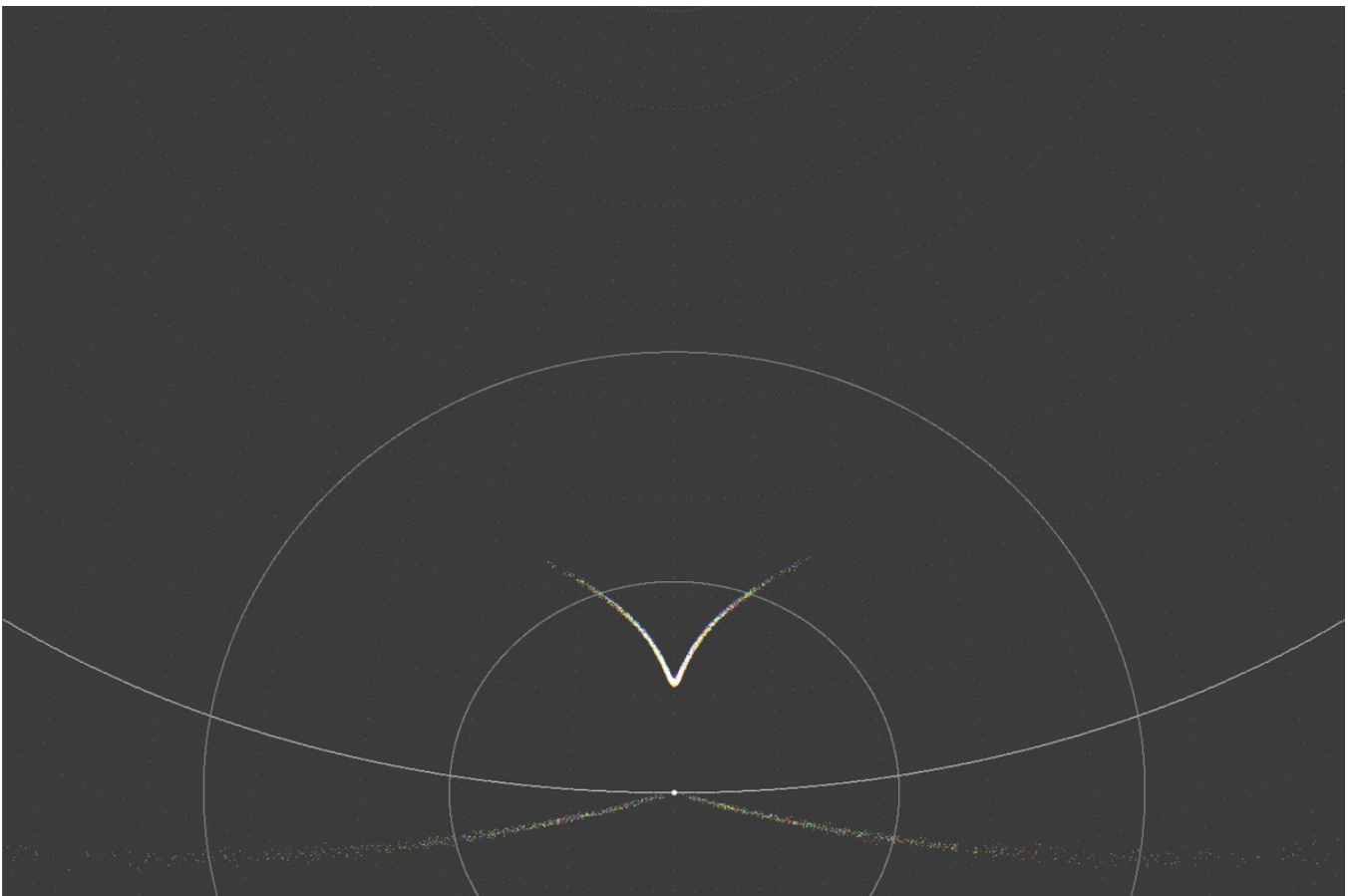






3/9 - 2026-02-08 09:25 UTC

7/8 Feb 2026, Rovaniemi. Moilanen arc and the related helix arc at horizon light.



4/9 - 2026-02-08 10:34 UTC

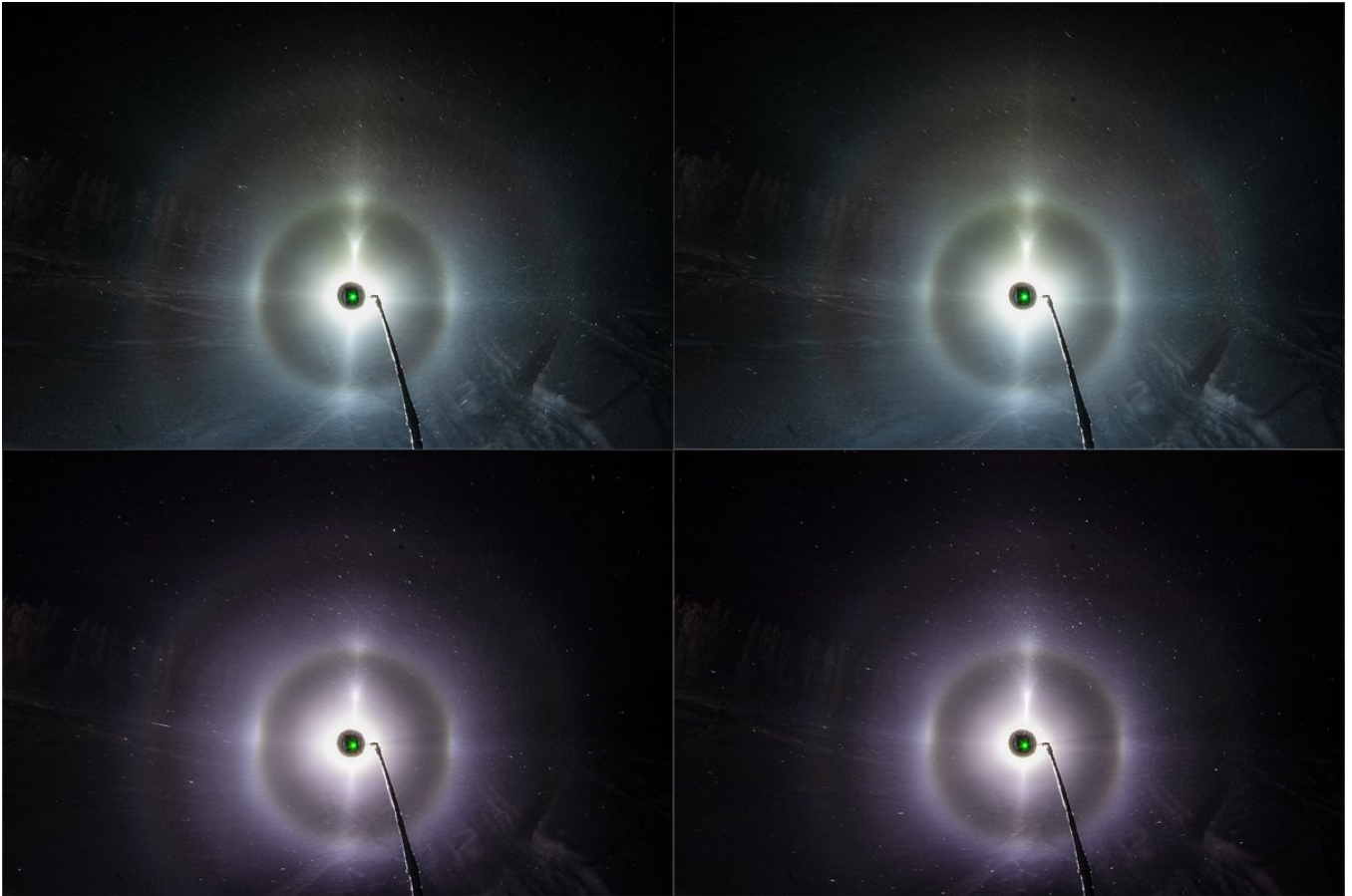
7/8 Feb 2026, Rovaniemi. Spotlight comparison for what it's worth. At one point I changed the HID (above) to Olight Javelot Turbo (below), not touching the camera settings. Comparison is hampered by the display seeming to have weakened during the change. Also the Olight beam does not seem to be as well centered as the HID beam.

Visually I prefer to be bathed in HID beam for its better immersion. The benefit of Olight is the smooth non-artifact beam. And of course you don't need to haul along a car battery, so it is unbeatable when you for example want to have the light on top of a steep mound.



5/9 - 2026-02-08 11:01 UTC

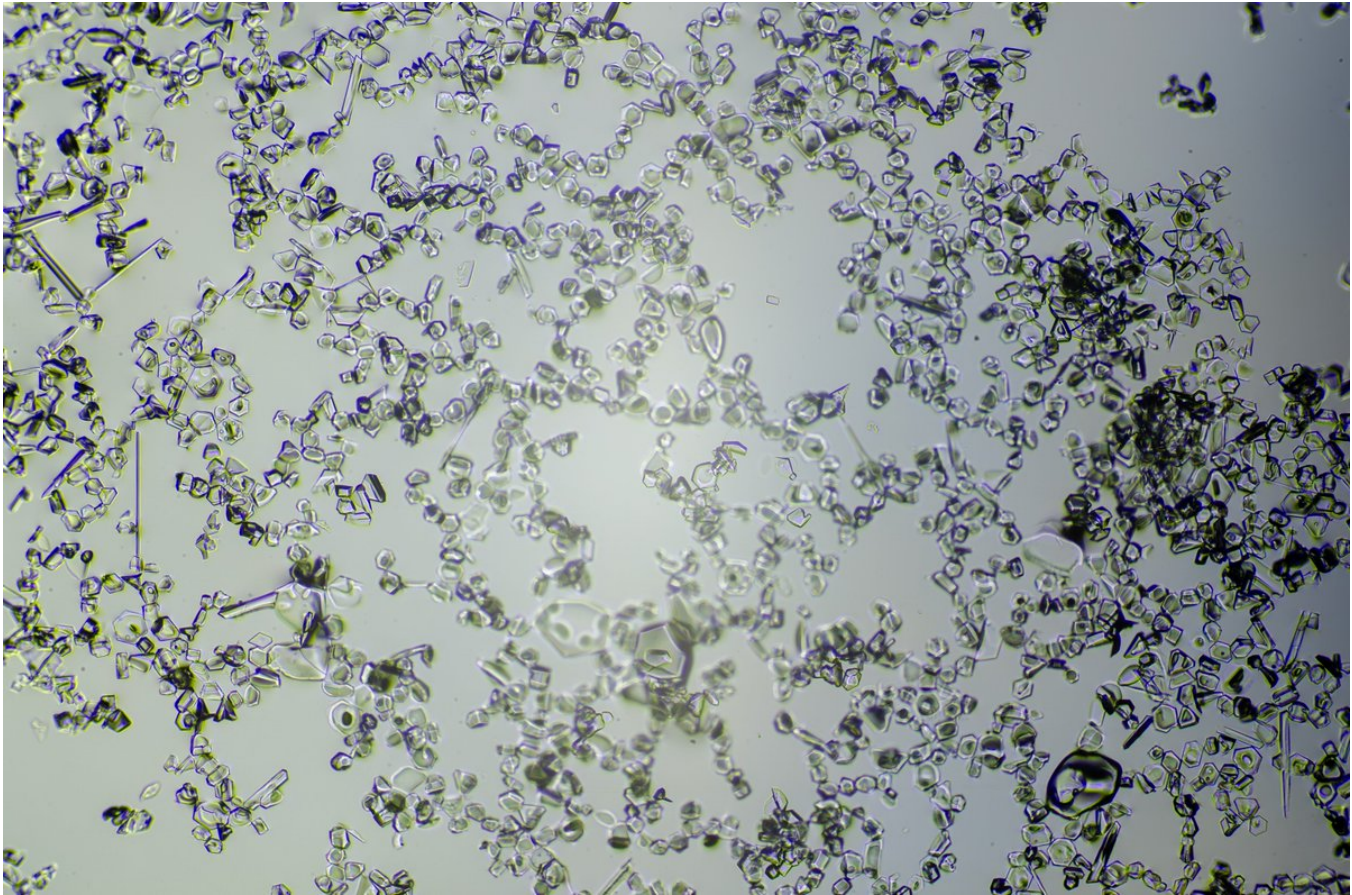
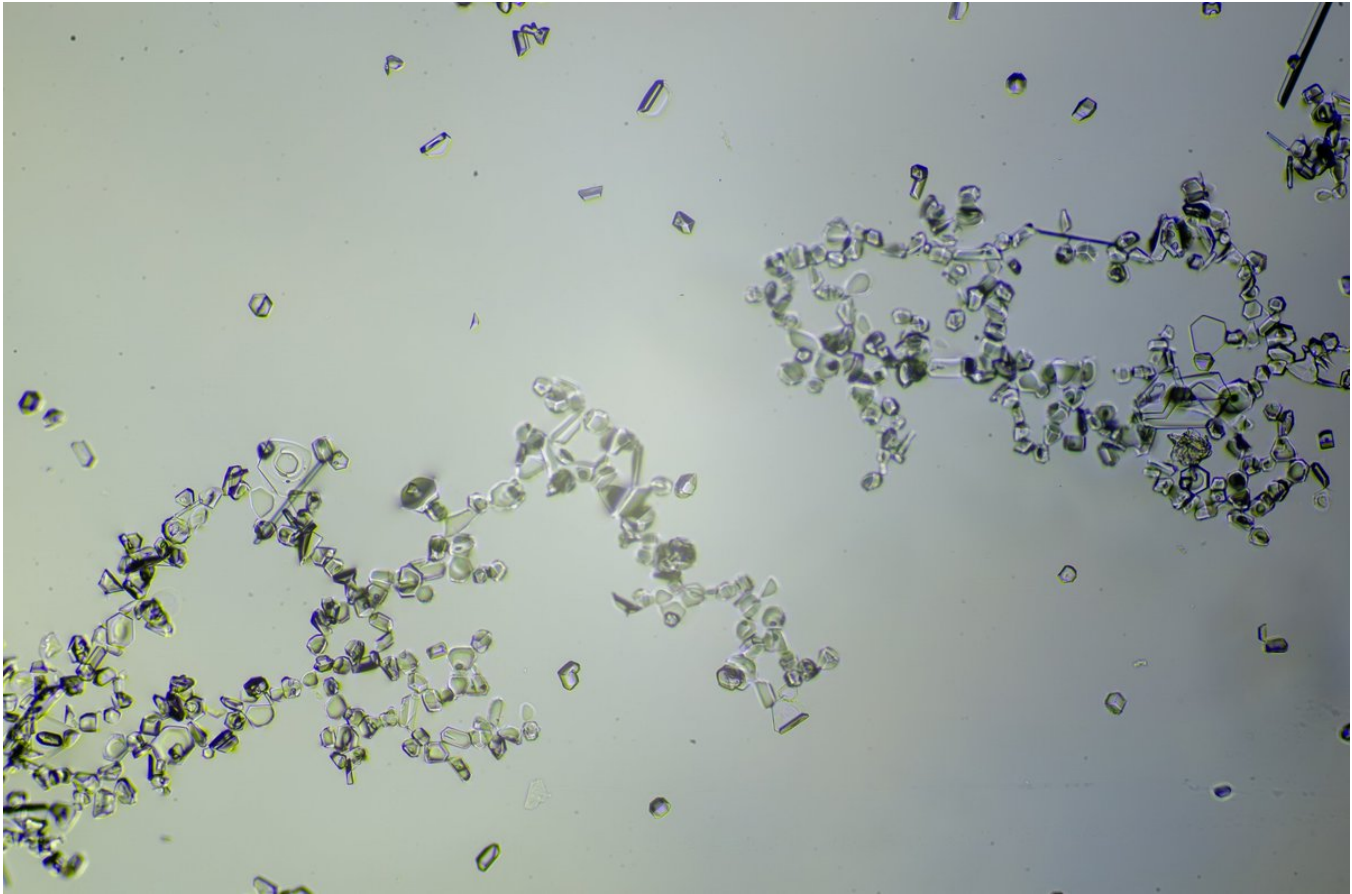
7/8 Feb 2026, Rovaniemi. Applied some camera raw adjustments in Photoshop for the collage.

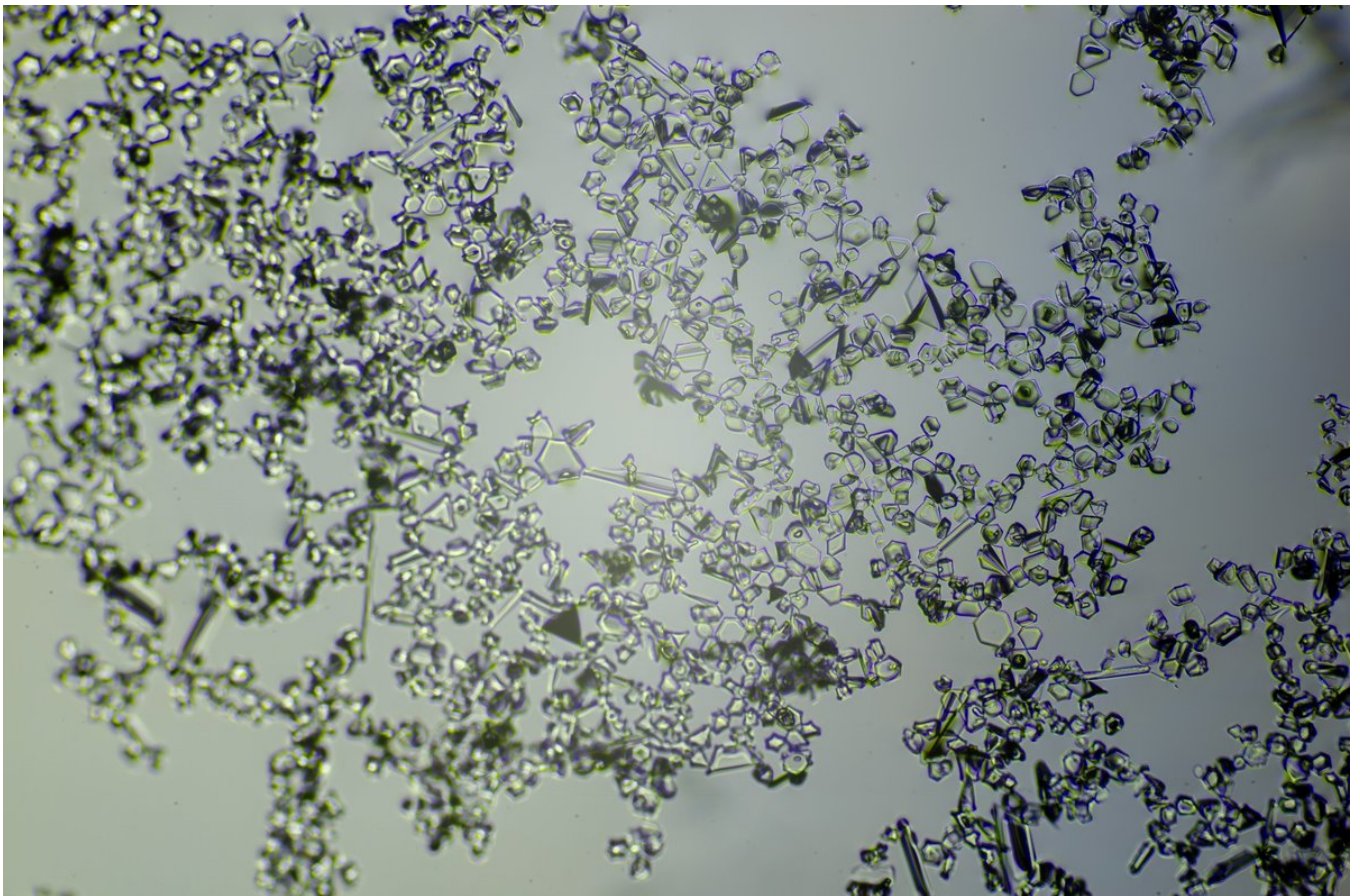
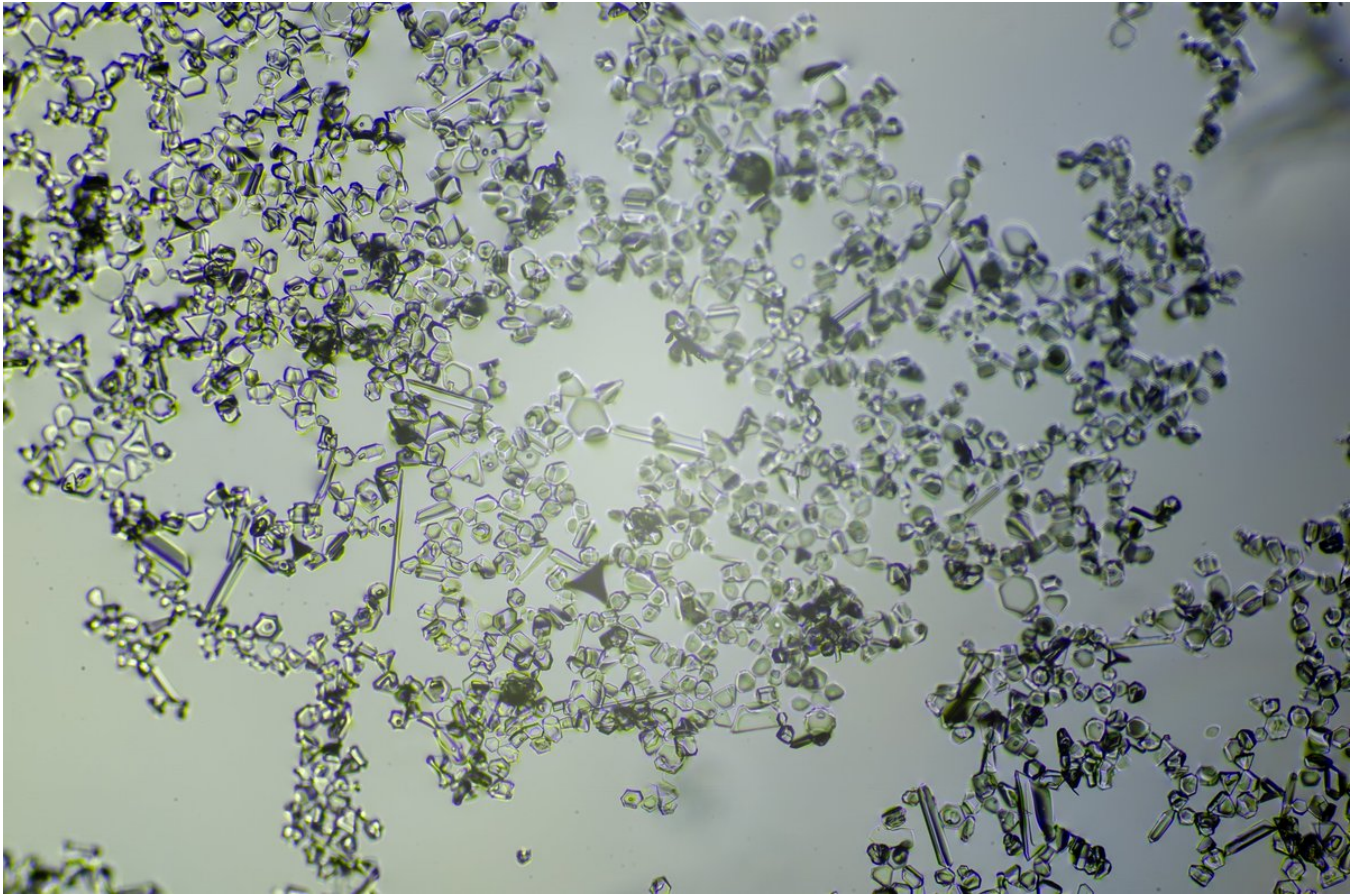


6/9 - 2026-02-08 11:58 UTC

7/8 Feb 2026, Rovaniemi. The microscope lighting issue indeed seems now to have been fixed, but something has been gnawing at the crystals. Equipment, including the liquid, was cold enough, so that leaves two possibilities: either the liquid has some alcohol in it or the led was too hot. This particular ignition fuel I use has its contents stated and they are all hydrocarbons, so nothing water soluble there. So it's the lamp then. But I wouldn't quite write the issue concluded that easy. Will do experimenting. Placed an order for purified gasoline which was unavailable in Finland earlier on in the winter.

As to this failed sample, it was taken at the beginning when the Moilanen arc was at its strongest.





7/9 - 2026-02-08 12:18 UTC

7/8 Feb 2026, Rovaniemi. Moilanen arc looked particularly good off-beam from a distance, this video

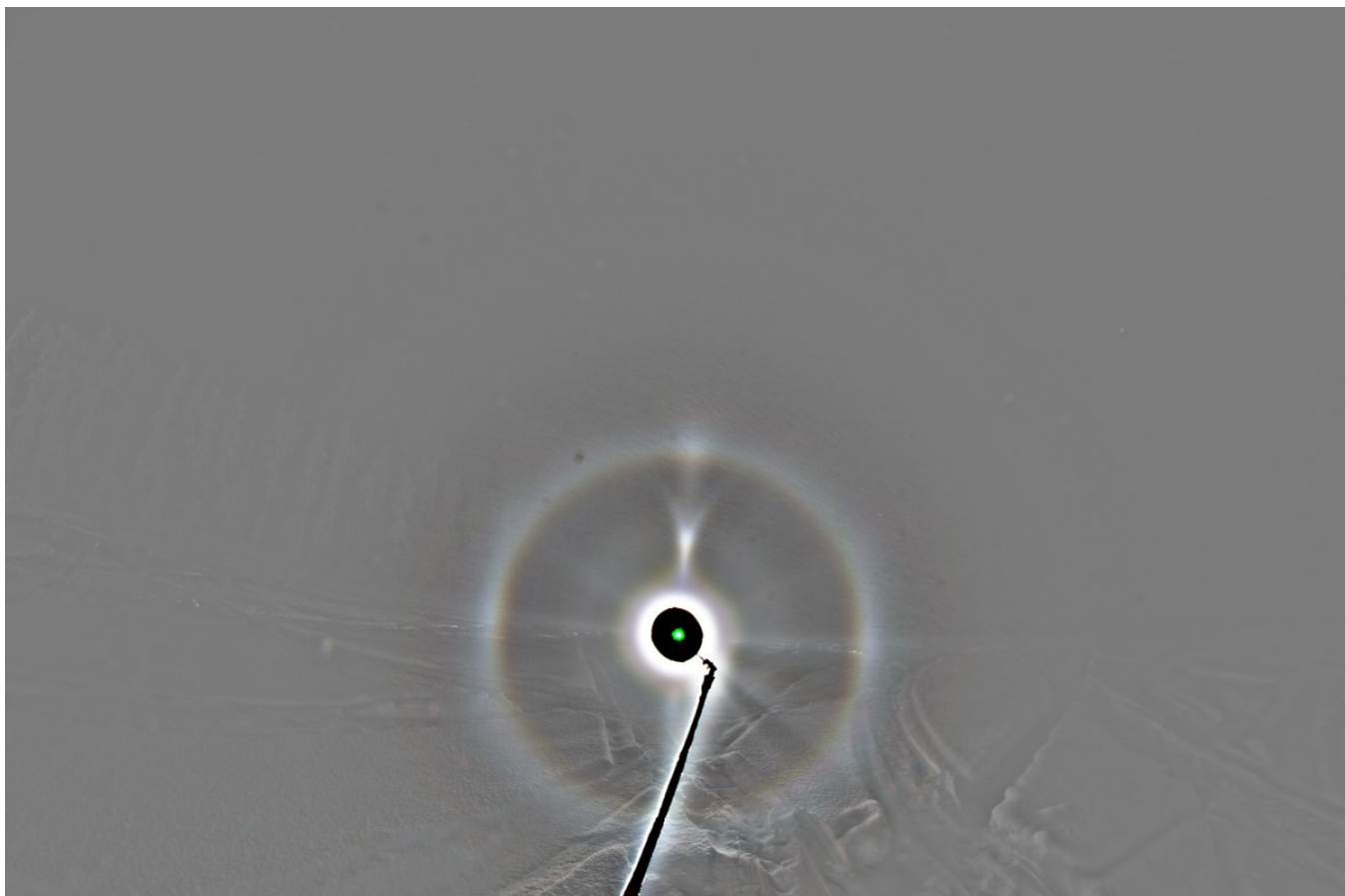
doesn't fully do justice to it.

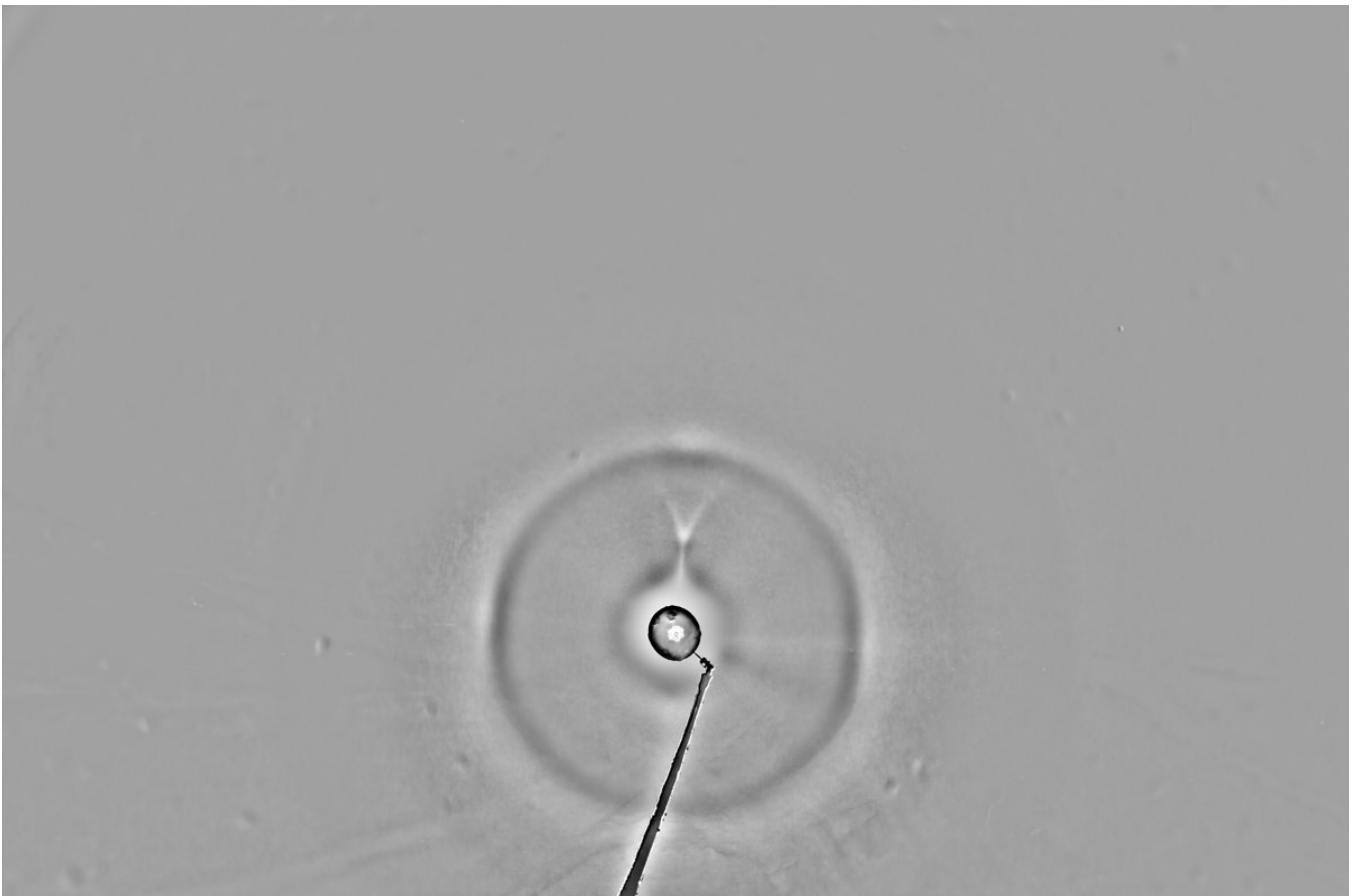
Original: <https://drive.google.com/file/d/1qRf2gc>

[video: 2020472184376820056-wHCE6o3sAO-2_cow.mp4]

8/9 - 2026-02-08 12:54 UTC

7/8 Feb 2026, Rovaniemi. 26 x 15s. Olight.





9/9 - 2026-02-08 13:28 UTC

7/8 Feb 2026, Rovaniemi. 29 x 15s, Olight. Half an hour after the previous post's stack. What's going

on outside 22° halo on the lower half? Must be artefacts given the contrast with upper half. Such a difference between the against-the-sky and against-the-snow parts of the display has been evident also in earlier displays. Possibly the issue concerns leds in particular, something to do with the narrower and more uneven spectrum compared to HID's. That's the summary I got when asked Grok. That's quite a bad downside of led, then.

The spectrum separation in parhelia looks also strange. In the intervening half hour I probably had Canon on the tripod. Will look at those later. I'll also do stacks where I changed from led to HID to see how much b-r and g-r differ.



